PARAS DESHPANDE

AI Product Engineer | GenAI | LLMOps | AI Automation | Product Analytics

Pune, India | +91 8788644033 | paraspdeshpande@gmail.com

linkedin.com/in/paras-deshpande-aiengineer • github.com/ParasDeshpande • parasdeshpande.github.io

PROFESSIONAL SUMMARY

AI Product Engineer with **1+ year** of experience designing, building and evaluating **production GenAI systems**. Skilled in **conversational AI, RAG, LLM evaluation**, AI product development, automation, **SQL analytics** and cross-functional product delivery. Experienced building AI products **from PRDs through implementation** while working with Product, Engineering, QA and Operations teams.

EXPERIENCE

Product Development Engineer | CMS, Pune *Aug 2026 – Present*

SOP Management Platform

- Owned **end-to-end delivery** of an **SOP management platform**, from stakeholder requirement gathering and process discovery through product design, development, testing and client delivery.
- Conducted **8–9 user interviews** and worked directly with stakeholders to map existing SOP workflows and translate operational requirements into product requirements, user flows and system specifications.
- Designed and developed workflows for SOP creation, **multi-stage approval** (up to three levels, approvers assigned per form), submission, tracking, reporting and audit management, with authentication and **role-based access** via NextAuth.js.
- Built the platform using **Next.js 14, TypeScript**, Tailwind CSS, Prisma, **PostgreSQL**, NextAuth.js, Node.js, **Docker** and React Native/Expo.
- Managed product iterations across requirement clarification, solution design, development, testing and delivery, resulting in an **MVP** that digitizes and standardizes SOP management.

AI Product Engineer (Apprentice) | Amber *Feb 2026 – Aug 2026*

AI Product Development

- **Led end-to-end development** of an AI-powered **Business Development Intelligence Platform** integrating CRM data with **LLMs** for reporting, prospect discovery, enrichment, AI email generation and outreach automation.
- Designed and shipped **five core modules**: Reporting Dashboard, AI Prospecting, Data Enrichment, AI Email Drafting and Outreach Sequencing.
- Ran **stakeholder and user interviews** with BD teams and turned their differing follow-up processes into one standardized flow: keyword prospect search, background fit check, first-contact draft and scheduled follow-ups.
- Defined **PRDs**, user journeys, technical specifications and system architecture for multiple AI initiatives.
- Built AI workflows using **OpenAI GPT**, SerpAPI, Apollo API, DuckDuckGo Search, Google Sheets and **Freshsales CRM**, validating API outputs with **Postman**.
- Designed **KPI dashboards** and reporting systems for Business Development teams.
- Redesigned the **AI property recommendation engine** with intelligent filter-relaxation logic; analyzed **6,000+ Mandarin search queries** to understand search-bar usage and proposed OpenSearch and AI re-ranking improvements.
- Authored specifications for Social Listening Platform, SOP Management Portal, WhatsApp Sales Agent, API Usage Tracking System and AI Gateway enhancements.

Conversational AI Evaluation & LLMOps

- Led quality evaluation of **Amber Copilot**, a chatbot supporting international student accommodation.
- **Prioritized** and shipped chatbot tonality, response-length, knowledge-base and message-template improvements based on failure analysis; against an **80% CSAT target (OKR)**, thumbs-up/down **CSAT rose from 30% to 60–65% over 4 months**.
- Evaluated **1,400+ conversations** to identify **hallucinations**, recommendation failures, booking loops, entity resolution issues and escalation gaps.
- Designed **LangSmith evaluation frameworks** with tracing, golden datasets, failure-mode datasets, journey taxonomy and automated quality metrics.
- Built **amber-eval-generator** with a 26-field evaluation schema and 14-node conversation classification pipeline.
- Analyzed LangSmith grader outputs to identify preference extraction failures, over-inference and entity resolution issues.
- Set up **LLM-as-judge** grading in LangSmith; grader reliability improved as the evaluation query set was expanded and refined.
- Implemented **guardrails** to prevent leakage of property-owner details and used **prompt versioning** to test prompts, responses and agent behavior.
- Ran a **50/50 A/B test** on a chatbot routing architecture change to compare how users responded, and ran **prompt experiments** with versioning.
- Developed **Python** data pipelines to transform SQL exports into aggregated failure analysis reports.

Engineering & Product Operations

- Authored developer-ready **Jira** tickets with reproduction steps, root-cause analysis and acceptance criteria.
- Wrote optimized **Amazon Redshift SQL** queries for reconstructing WhatsApp conversations and generating evaluation datasets.

- Standardized chatbot fallback messaging for consistency and customer experience.
- Collaborated with Product, QA, Engineering and Operations in **Agile/Scrum** cycles (sprint planning, daily standups) to validate AI releases through **UAT**.
- Built a Book Form Automation PoC reducing manual effort by **4–8 minutes** per interaction.
- Produced SQL, **Metabase**, Excel and Notion reports supporting product releases and executive decision-making.

Digital Operations Consultant (Freelance) | Bakery Manufacturing Business, Pune *Sep 2025 – Feb 2026*

- Digitized the product catalog of a **100+ SKU** bakery manufacturer supplying retailers across Pune.
- Built a **Google Sheets workflow** for production and sales tracking, giving the owner visibility into output, sales and best-performing products.
- Standardized customer order messaging and introduced **AI tools** for basic analysis and content creation.
- Worked with the owner on **packaging, marketing and pricing** for bakery products.

Data Scientist Intern | Geta.AI Labs Pvt. Ltd. *Jun 2025 – Sep 2025*

- Engineered **multi-LLM RAG pipelines** (OpenAI, Claude, Gemini, LLaMA) with **semantic chunking and indexing** to improve retrieval on hard-case questions; compared model outputs across providers on quality and budget.
- Developed production-grade **conversational AI** systems using **LangChain**, ChromaDB, PostgreSQL, Redis and MongoDB.
- Built **hybrid retrieval** combining **PostgreSQL** (structured university data) with **ChromaDB** (PDF documents); worked with **FAISS** and evaluated **Pinecone**.
- Optimized **cost and response latency** by reducing the context sent to the LLM through semantic chunking and indexing.
- Built a **FastAPI + Streamlit** architecture with automated logging.
- Integrated external APIs for scalable, context-aware AI applications, checking API responses with **Postman**.
- Worked on modular backend architecture, retrieval optimization, prompt engineering and deployment workflows.

PROJECTS

PCL — Project Coordination Log | Product / Full-Stack Project

- Designed and developed a **project coordination platform** to centralize project activities, tracking, communication and operational visibility.
- Built using **Next.js 14, TypeScript**, Tailwind CSS, **Python FastAPI**, SQLAlchemy, SQLite and Recharts.
- Developed **dashboards** and structured workflows for project tracking, coordination and reporting.
- Designed the architecture with a **PostgreSQL-ready backend** for future production scaling.

Enterprise RAG Knowledge Assistant (personal project): Semantic retrieval system with PDF ingestion, embeddings and grounded answers to reduce **hallucinations**.

AI Business Analytics Platform (personal project): Churn prediction, sentiment analysis and sales forecasting models built to demonstrate core **ML fundamentals**.

Sales Insights Dashboard (personal project): Power BI dashboards with sales forecasting and performance analytics.

TECHNICAL SKILLS

AI / LLMs: OpenAI GPT, Claude, Gemini, LangChain, LangGraph, LangSmith (tracing & evaluation), RAG, **Vector Databases** (FAISS, ChromaDB, Pinecone), Embeddings, AI Agents, Prompt Engineering & Versioning, **LLM-as-Judge**, LLM Evaluation, Guardrails

Languages & Data: Python, SQL, TypeScript, pandas, NumPy, scikit-learn, ETL / Data Pipelines

Backend & Frontend: FastAPI, REST APIs, API Integration & Testing (**Postman**), Node.js, Next.js, React Native, Authentication & Role-Based Access

Databases: PostgreSQL, MongoDB, Redis, Amazon Redshift, Prisma

DevOps & Cloud: Git / GitHub, Docker, Vercel, Render, AWS (basics), Linux, Manual Testing

Analytics & Tools: Power BI, Metabase, Excel, Google Sheets, Data Visualization, CSAT & Customer Feedback Analysis, n8n, Make, Jira, Notion, Freshsales

Product & Process: PRDs, User Stories, Roadmaps, User Flows, Prioritization, **OKRs**, Experimentation, Go-to-Market (packaging, marketing, pricing), **User Research**, User Interviews, **Agile / Scrum**, Sprint Planning, UAT, A/B Testing, Funnel & Adoption Metrics, KPI Dashboards, System Design, Documentation

EDUCATION

B.E. Artificial Intelligence & Machine Learning, Savitribai Phule Pune University | CGPA: 8.0 *2025*

PUBLICATION

International Journal of Data Science and Analytics (2025)